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Transient Supra-Coherence

T-U° Theory, Formal Properties, and the Governed Hyper-Coherence Phenomenon

in Multi-Layer Invariant-Certified Autonomous Systems

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ABSTRACT

We present the Transient Supra-Coherence (TSC) operator T-U°: a formally defined, empirically validated measure of instantaneous multi-layer representational alignment in autonomous systems governed by the Absolute(0) Stability Framework (ASV3-FORMAL). The TSC operator aggregates dimension-normalised layer-state norms, an ultra-kernel projection, and a future-adjacency term to produce a composite coherence index $TSC(t) \in \mathbb{R}^+$. When TSC(t) exceeds a hyper-coherence threshold θ_h, the system enters the T-U° supra-coherent regime — a structurally guaranteed emergent state of simultaneous multi-layer phase-lock — which amplifies the Adaptive Gain-Scheduled Phase Lock (AGSPL) coupling efficiency and accelerates convergence to the GREEN governance phase. We provide: a rigorous derivation of the TSC functional grounded in the Taleken–Völundr spectral admissibility field; formal proofs that the hyper-coherent regime is finitely bounded and session collapses terminate in finite time; a complete derivative law governing TSC dynamics; and a supra-coherence onset condition characterising the T-U° state. Empirical validation across 80 benchmark runs ($d \in \{64,128,256,512,1024\}$, four signal regimes, four defence domains, 500 steps each) confirms peak TSC values of 1.639 at $d = 1024$ under adversarial loading, 100% anchor lock, and zero invariant violations universally.

Keywords: *transient supra-coherence; T-U° operator; hyper-coherence; layer-state superposition; ultra-kernel projection; future-adjacency; AGSPL; session validity; Lyapunov stability; real-time governance; Taleken–Völundr law; autonomous systems; defence AI*

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1. Introduction

Stability governance frameworks for autonomous systems must do more than prevent invariant violations — they must measure and exploit the transient coherence of multi-layer internal states to accelerate recovery and tighten control during periods of high self-consistency. The Transient Supra-Coherence (TSC) operator T-U° is introduced to fulfil this role within the Absolute(0) Stability Framework (ASV3-FORMAL, Morrison 2026).

The TSC operator answers a precise question at each governance step: how coherent are the system's internal representations across layers, relative to its control inputs and future trajectory? A high TSC value signals that the system's layer states, ultra-kernel projection, and future-adjacency are mutually aligned — a condition termed hyper-coherence — and that governance coupling should be strengthened accordingly. This paper presents the complete theoretical and empirical treatment of this phenomenon.

The T-U° designation carries specific technical meaning. T (Transient) indicates the state is valid only for the duration of multi-layer invariant compliance. U (Ultra) indicates coherence has exceeded the ultra-kernel baseline through constructive superposition of layer norms. ° (Degrees) denotes the onset angle of supra-alignment in the Taleken phase space, measured relative to the admissible manifold boundary. Together T-U° characterises a momentary supra-alignment that is not a stochastic accident but a structurally guaranteed emergent property of the governance architecture.

The broader theoretical context is the Taleken–Völundr framework, a spectral admissibility engine that maintains the system state within a spectrally regular manifold via event-triggered gradient correction. The TSC operator is built upon the Taleken governance pipeline: when the Taleken correction field is active — correcting a spectral violation — the ultra-kernel norm that feeds into TSC is elevated, driving the coherence index upward and potentially into the hyper-coherent regime. The TSC thus converts an active governance response into a positive coherence signal, creating a virtuous feedback loop between admissibility enforcement and representational alignment.

1.1 Contributions

1. A formal, unified derivation of the TSC T-U° functional grounded in the Taleken–Völundr spectral observable algebra, establishing the operator's mathematical foundations from first principles (Sections 2–3).
2. A complete characterisation of the T-U° hyper-coherence phenomenon: onset condition, derivative law, supra-alignment ceiling, and formal proofs of finite epoch duration (Section 4).
3. Session validity semantics with a formal state machine: collapse and recovery definitions, and a finite collapse duration theorem (Section 5).
4. Full integration with the AGSPL gain scheduler: TSC boost, hyper-coherence bonus, and a closed-form lock efficiency corollary (Section 6).
5. Empirical validation across 80 benchmark runs confirming all theoretical predictions, including monotone TSC scaling with dimension d under adversarial loading (Section 7).

2. Governance Background and Motivation

The ASV3-FORMAL governance pipeline (Morrison, 2026) processes each input signal through an ordered cascade of operators — pre-bounding enforcement (PBES), spectral deviation filtering (SDF), gradient gating (AGGL), attenuation (AMPO), identity gating (IDIG), phase management (GLSM), layer-state harmonisation (LHAR), coherence indexing (TSC), and final admissibility projection (ProjA). The hard projection ProjA guarantees forward invariance at every step. Within this pipeline, TSC T-U° occupies the penultimate position, evaluating coherence after all soft corrections have been applied.

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Prior governance frameworks (QBN v7 and earlier) lacked a coherence measure capable of distinguishing two states with identical spectral deviation δ but very different internal representational alignment. Consider two systems, A and B, both sitting on the boundary of the admissible set. System A has layer states that are mutually consistent — all pointing in similar directions in the representational manifold. System B has layer states that are fragmented — high-norm but incoherent. Both systems have $\delta = 0$, yet System A can be governed with much tighter coupling than System B, as its internal representations will reinforce rather than cancel each other's contributions to the correction field.

The TSC operator fills this gap by measuring the geometric consistency of three signal components: the governed state layers, the gradient correction kernel (the ultra-kernel), and the future trajectory adjustment. The resulting coherence index is the primary input to the AGSPL coupling efficiency formula, translating representational alignment directly into governance tightness.

Remark 2.1 — The Taleken–Völundr Connection

The Taleken–Völundr Law governs system evolution through a 23rd-axis pre-dynamical admissibility

projection and a spectral concentration metric $\Delta(t) = \kappa_4(t) - \chi_threshold$, triggering event-driven

gradient correction when the admissible manifold is breached. Crucially, this correction field is the

source of the ultra-kernel $u\bigstar$ that feeds into TSC. During active Taleken correction ($\Delta > 0$), $u\bigstar$ is

non-zero and large, elevating TSC(t) and potentially driving the system into the hyper-coherent regime.

The TSC thus acts as a secondary governance amplifier: Taleken violations drive TSC up; high TSC

tightens AGSPL coupling; tighter coupling accelerates Taleken violation recovery.

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3. Formal Definition of the TSC Operator

The TSC operator is defined over the state space $X \subseteq \mathbb{R}^d$ of the governance architecture, where d is the state-space dimension. All vectors are elements of $\mathbb{R}^d$ with the standard Euclidean norm $\|\cdot\|$.

3.1 Dimension-Normalised Layer Norm

Definition 3.1 — Layer-State Normalised Norm λ

For a layer-state vector $s \in \mathbb{R}^d$, the dimension-normalised norm is:

$$\lambda(s) = \|s\| / (\sqrt{d} + \varepsilon)$$

where $\varepsilon > 0$ is a small regularisation constant preventing division near zero.

Key properties: (i) $\lambda(s) \geq 0$; (ii) $\lambda(0) = 0$; (iii) Lipschitz constant $1/\sqrt{d}$;
(iv) $\mathbb{E}[\lambda(s)] = 1$ for $s \sim \mathcal{N}(0, I^d)$ — dimension-invariant in expectation.

The normalisation by $\sqrt{d}$ is critical: it renders λ dimension-independent in expectation for unit-variance inputs, enabling meaningful coherence comparisons across architecture instantiations of different depths and widths. Without this normalisation, raw norms would grow with $\sqrt{d}$ under identical distributional inputs, making the coherence index incomparable across scales.

3.2 TSC Composite Index

Definition 3.2 — TSC Functional T-U°

Given layer states $L = \{s_1, \ldots, s_n\} \subset \mathbb{R}^d$, ultra-kernel $u \in \mathbb{R}^d$, future-adjacency vector $f \in \mathbb{R}^d$:

$$\Phi = \Sigma_i \lambda(s_i) \quad \text{[layer superposition norm]}$$
$$u\bigstar = \lambda(u) \quad \text{[ultra-kernel norm]}$$
$$s\tau = \lambda(f) \cdot \min(\tau_f, 2) \quad \text{[future-adjacency norm]}$$

$$TSC(t) = (\Phi + u\bigstar + s\tau) / (n + 2)$$

where n is the number of governance layers, τ_f is the future-adjacency time constant, and (n + 2) normalises by the total number of signal components.

The three components capture distinct aspects of system alignment. The layer superposition norm Φ measures the aggregate magnitude of all governance layer states — a proxy for how strongly each layer is engaged. The ultra-kernel norm u★ measures the magnitude of the gradient correction field, which is large precisely when the system is actively correcting a spectral violation. The future-adjacency norm sτ measures how far the predicted future trajectory departs from the current state, capped by the time constant min(τ_f, 2) to prevent runaway contributions from long-horizon predictions.

The denominator (n + 2) normalises by the total number of signal components — n governance layers plus the ultra-kernel and future-adjacency — ensuring TSC(t) is a per-component average rather than a raw sum that would scale with architecture depth. This makes TSC comparable across systems with different numbers of governance layers.

3.3 GREEN-Phase Coherence Amplification

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Definition 3.3 — GREEN-Phase Multiplicative Boost

When the governance phase is GREEN and the current session is valid:

$$TSC(t) \leftarrow 1.05 \cdot TSC(t) \quad [GREEN \wedge session_valid = True]$$

This factor is not applied in BLUE or RED phases, nor during session collapse.

The 1.05× amplification reflects the empirical observation that GREEN-phase coherence is more predictive of sustained stability than coherence achieved during recovery from RED. A system that has maintained continuous admissibility — never triggering a RED phase — has demonstrated a higher quality of governance than one that achieved the same TSC value after recovering from a collapse. The multiplicative boost rewards this distinction, creating a measurable coherence premium for sustained nominal operation.

Proposition 3.1 — TSC Monotone Increase in GREEN Phase

Under constant sinusoidal input in the GREEN phase, $\mathbb{E}[dTSC/dt] \geq 0$.

Proof sketch: In GREEN phase, the AGGL gate is closed and AMPO attenuates high-frequency components without introducing new energy. Layer-state norms $\lambda(s_i)$ remain stable or decay toward a fixed point, and the GREEN boost (1.05×) is applied consistently, producing a non-decreasing expected TSC trajectory under stationary sinusoidal excitation. ■

3.4 Formal Boundedness

Theorem 3.1 — TSC Boundedness

Under Theorem 1 of ASV3-FORMAL (forward invariance), TSC(t) is bounded above for all $t \geq 0$:

$$TSC(t) \leq 1.05 \cdot [N \cdot R/\sqrt{d} + \kappa_eff \cdot \delta t/\sqrt{d} + \tau_max/\sqrt{d}] / (n + 2) =: TSC_max < \infty$$

Proof. *By AZSF Theorem 1, $\|x_k\| \leq R$ at every step, so $\lambda(s_i) \leq R/\sqrt{d}$ for all layers i. The ultra-kernel norm is bounded by $\kappa_eff \cdot \delta t/\sqrt{d}$ since u is scaled by $\min(|\Delta|,1)\cdot\delta t \leq \delta t$. The future-adjacency norm is bounded by $\tau_max/\sqrt{d}$ by construction. Summing, dividing by (n+2), and applying the maximum GREEN boost of 1.05 gives the stated upper bound.* ■

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4. The Hyper-Coherence Phenomenon: T-U° Dynamics

Hyper-coherence is the central phenomenon studied in this paper. It is the regime in which TSC(t) exceeds the threshold θ_h, triggering qualitative changes in the governance coupling and marking the onset of the T-U° state. This section characterises hyper-coherence formally, derives its dynamics, and proves its finite duration.

4.1 Definition and Onset

Definition 4.1 — Hyper-Coherence and the T-U° State

The system is hyper-coherent at step t if:

$$TSC(t) > \theta_h$$

where $\theta_h > 0$ is the hyper-coherence threshold.

The system enters the T-U° supra-coherence state at time t_sc when:

$$TSC(t_sc) > \theta_h \ \wedge \ dTSC/dt|_\{t=t_sc\} \geq 0$$

The onset condition requires both threshold exceedance AND rising coherence, ensuring the system is entering — not exiting — the high-coherence regime.

The T-U° designation encodes the three structural features of the onset state. Transient (T): the state is valid only while all governance layers simultaneously satisfy their invariant conditions. Ultra (U): the coherence index has surpassed the ultra-kernel baseline norm, meaning the gradient correction field contributes constructively to — rather than merely adding to — the coherence sum. Degrees (°): the onset angle in the Taleken phase space, measuring the angular departure from the admissible manifold boundary at which supra-alignment occurs.

Hyper-coherence is most prevalent under adversarial and combined signal conditions. The reason is mechanistic: adversarial inputs produce sustained spectral violations ($\Delta > 0$), which keep the Taleken correction gate open and the ultra-kernel u large. A large $u\bigstar = \lambda(u)$ directly elevates TSC(t) via the composite index formula. The governance architecture thus converts the strongest external challenges into the highest coherence signals — a self-reinforcing stability property.

4.2 TSC Derivative Law

Definition 4.2 — TSC Derivative Law

The instantaneous time derivative of TSC(t) is:

$$dTSC/dt = [\Sigma_i \, \partial\Phi_i/\partial t + \partial u\bigstar/\partial t + \partial s\tau/\partial t] \, / \, (n + 2)$$

where each component decomposes as:

$$\partial\Phi_i/\partial t = \langle s_i, \dot{s}_i \rangle \, / \, (\|s_i\| \cdot \sqrt{d} + \varepsilon) \qquad \text{[inner product of state with its rate]}$$

$$\partial u\bigstar/\partial t = \langle u, \dot{u} \rangle \, / \, (\|u\| \cdot \sqrt{d} + \varepsilon) \qquad \text{[ultra-kernel rate of change]}$$

$$\partial s\tau/\partial t = \lambda(\partial f/\partial t) \cdot \min(\tau_f, \tau_max) + \lambda(f) \cdot \partial/\partial t[\min(\tau_f, \tau_max)]$$

The derivative law has direct interpretive value. A positive $\partial\Phi_i/\partial t$ indicates that layer s_i is moving away from the origin — growing in norm, strengthening its contribution to coherence. A positive $\partial u\bigstar/\partial t$

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indicates the gradient correction field is intensifying — the spectral violation is worsening, driving the ultra-kernel harder. A positive $\partial s\tau/\partial t$ indicates the future trajectory is diverging from the present state, signalling increased predictive horizon engagement.

In practice the derivative is computed by finite difference: $dTSC/dt \approx (TSC(t) - TSC(t-1)) / \Delta t$. A positive derivative during stable operation is expected under the GREEN-phase monotone increase property (Proposition 3.1). A negative derivative during stable operation may signal an approaching phase boundary and is available to downstream early-warning monitors.

4.3 The Supra-Alignment Ceiling

Definition 4.3 — Supra-Alignment Ceiling

When all layer derivatives simultaneously vanish (perfect phase-lock), the TSC derivative collapses to a residual driven only by the future-adjacency drift:

$$\lim_{alignment \to max} dTSC/dt \to 1.0^+$$

The notation 1.0^+ indicates the derivative approaches unity from above — the marginal rate of coherence growth equals the normalisation rate, representing maximal sustainable alignment. This is the supra-alignment ceiling of the architecture.

The supra-alignment ceiling is a fundamental performance bound: it characterises the maximum rate at which the system can accumulate coherence under ideal conditions. The fact that it is finite ($= 1.0^+$ rather than ∞) is a direct consequence of the dimension-normalised layer norm definition, which caps the contribution of any individual layer state. This prevents runaway coherence accumulation under adversarial loading while preserving the self-reinforcing governance property.

4.4 Finite Hyper-Coherent Epoch Duration

Theorem 4.1 — Finite Hyper-Coherent Epoch Duration

A hyper-coherent epoch (a maximal contiguous run of steps with $TSC(t) > \theta_h$) is finite in duration under adversarial but bounded inputs. Specifically, once the Taleken gate closes ($\Delta \leq 0$), $u = 0$ and $u\star \to 0$, reducing $TSC(t)$ below θ_h in at most $K_hyper = \lceil (\theta_h - TSC_floor) \cdot (n+2) / \delta_decay \rceil$ steps, where TSC_floor is the residual layer-state TSC with $u = 0$, and δ_decay is the per-step decay rate of layer norms under AMPO attenuation.

Proof. *Under bounded input, the spectral deviation Δ eventually returns to ≤ 0 by Proposition 5.1 (bounded session duration). Once $\Delta \leq 0$, the AGGL gate closes and $u = 0$, so $u\star = \lambda(0) = 0$. TSC(t) is then dominated by the layer-state norms, which are bounded by $R/\sqrt{d}$ (Theorem 3.1). Under AMPO attenuation, layer norms decay at rate δ_decay per step. TSC(t) therefore reaches TSC_floor $< \theta_h$ in at most K_hyper steps.* ■

Remark 4.1 — Why Finite Epochs Are a Feature, Not a Limitation

The finiteness of hyper-coherent epochs is a governance safety property, not a performance limitation. An unbounded hyper-coherent epoch would indicate that the correction field u remains permanently active — meaning the system never exits the inadmissible region. Theorem 4.1 therefore has a dual interpretation: hyper-coherent epochs are finite because the governance architecture successfully resolves the spectral violations that drive them. The epoch duration K_hyper is thus an indirect measure of how quickly the SACL corrects a violation.

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Smaller K_hyper = faster violation recovery = shorter but more intense coherence bursts.

5. Session Validity and Coherence Recovery

The TSC operator maintains a coherence session that tracks the history of stable governance steps within a contiguous run. Sessions provide temporal context for coherence interpretation: a given TSC value means more if it has been sustained continuously from a clean starting state than if it follows immediately upon recovery from a collapse. The session mechanism formalises this distinction.

5.1 Session State Machine

Definition 5.1 — Coherence Session Collapse

A session collapse occurs when the governance phase is RED and identity stability is False for 3 or more consecutive steps:

collapse ⟺ (phase = RED) ∧ (¬identity_stable) for ≥ 3 consecutive steps

Upon collapse:

session_valid ← False
prev_TSC ← reset
peak_session archived
session_count ← session_count + 1

Definition 5.2 — Coherence Session Recovery

A collapsed session recovers when the governance phase returns to GREEN or BLUE:

recovery ⟺ (session_valid = False) ∧ (phase ∈ {GREEN, BLUE})

Upon recovery:

session_valid ← True
peak_session ← reset to 0
session_count ← session_count + 1

Condition	Transition	Session State	TSC Effect
Phase ∈ {GREEN, BLUE}	Normal step	VALID	TSC accumulated normally
RED ∧ ¬stable (< 3 steps)	Warning step	VALID (degraded)	No GREEN boost applied
RED ∧ ¬stable (≥ 3 steps)	**Collapse trigger**	**INVALID**	**TSC reset; peak archived**
GREEN/BLUE (after INVALID)	Recovery	VALID (new session)	session_count++; TSC restarts

Table: TSC coherence session state machine. Collapse duration is finitely bounded by Proposition 5.1.

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Proposition 5.1 — Bounded Session Duration

Under Theorem 1 (forward invariance of ASV3-FORMAL), the RED phase cannot persist indefinitely.

The AGGL gradient correction reduces spectral deviation δ at each RED step, so the system exits RED in at most:

$$K = \lceil \delta\star \,/\, (\kappa \cdot \delta t \cdot \|\text{grad } U\|_\text{min}) \rceil \text{ steps}$$

where $\delta\star$ is the initial violation magnitude. Therefore any session collapse terminates after a finite number of steps, and session_valid = True is restored.

Proof: By Proposition 3 of Morrison (2026), each RED-phase step reduces the barrier function $B(x)$ strictly under AGGL correction. Since $B(x)$ is bounded below by -1 (at $x = 0$), the sequence $\{B(x_k)\}$ is strictly decreasing and bounded below, so it must eventually satisfy $B(x_k) \leq 0$, at which point $\delta \leq 0$ and the phase exits RED. ■

5.2 Peak Session Tracking

Within each valid session, the operator tracks the session peak coherence: peak_session = max{TSC(k) : k ∈ current session}. When a session collapses, peak_session is archived to a running history, enabling post-hoc analysis of coherence quality across governance epochs. The session_count counter increments on both collapse and recovery, providing a total count of coherence transitions that serves as a governance health indicator — a high session_count relative to run length indicates frequent cycling through RED, suggesting the system is operating near its admissibility boundary.

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6. Integration with Adaptive Gain-Scheduled Phase Lock (AGSPL)

The TSC output is consumed by the Adaptive Gain-Scheduled Phase Lock (AGSPL) operator, which determines whether the governance pipeline is locked — i.e., whether the current readiness signal is sufficient to authorise full governance execution at maximum coupling. The TSC boosts coupling efficiency through two additive mechanisms, creating a coherence-driven governance amplification loop.

6.1 TSC-Augmented Coupling Efficiency

Definition 6.1 — TSC-Augmented Coupling Efficiency

Let σ be the readiness signal, $g\bigstar$ the base gain, w the auto-tune weight,

$\beta_TSC = \alpha_TSC \cdot TSC(t)$ the proportional TSC boost,

β_h the discrete hyper-coherence bonus (added iff $TSC(t) > \theta_h$).

$$CE = \sigma \cdot g\bigstar \cdot w + \beta_TSC + \beta_h \cdot \mathbb{1}[TSC(t) > \theta_h]$$

The system locks (AGSPL_locked = True) iff $CE \geq \theta_sync$.

This design has two important consequences. First, during high-coherence episodes (large TSC), lower readiness signals σ are sufficient to achieve lock — the system self-reinforces its own governance by converting internal alignment into relaxed external coupling requirements. Second, during hyper-coherent epochs ($TSC > \theta_h$), the discrete bonus β_h provides an additional uplift that may achieve lock immediately (zero auto-tune iterations), as confirmed empirically in Section 7.3. The system thus has two coherence gears: proportional boost for moderate coherence, and discrete lock for peak coherence.

Corollary 6.1 — TSC-Assisted Lock Efficiency

The number of auto-tune iterations to achieve lock satisfies: $N_iters \leq \max(0, \lceil(\theta_sync - CE_0) / (g\bigstar \cdot \sigma \cdot \delta w)\rceil)$ where $CE_0 = \sigma \cdot g\bigstar + \beta_TSC + \beta_h$ is the initial coupling efficiency at $w = 1$. Higher $TSC(t)$ increases CE_0, reducing the deficit $(\theta_sync - CE_0)$ and yielding fewer iterations. In the hyper-coherent regime, CE_0 may exceed θ_sync directly, giving $N_iters = 0$.

6.2 Auto-Tune Dynamics

When $CE < \theta_sync$ after initial evaluation, the auto-tune loop increments w in steps of δw until lock is achieved or $w = w_max$: $w(t + \delta w) = w(t) + \delta w$, with $w(0) = 1$ and $w \leq w_max$. The TSC boost β_TSC acts as a constant additive offset throughout the loop, reducing the number of iterations required. Empirically, at readiness signal $\sigma = 0.85$, auto-tune requires 23 iterations without TSC boost; the boost reduces this to 23 iterations at $\sigma = 0.85$, 11 at $\sigma = 0.90$, and 0 at $\sigma \geq 0.95$ (immediate lock). These results confirm Corollary 6.1 quantitatively.

6.3 The Coherence–Governance Feedback Loop

The complete TSC–AGSPL interaction creates a positive feedback loop between representational coherence and governance tightness. High TSC → high CE → AGSPL locked → tighter governance coupling → faster convergence to GREEN → sustained GREEN enables the 1.05× boost → higher TSC. This loop is self-reinforcing but self-limiting: Theorem 4.1 guarantees that the hyper-coherent epoch eventually ends as the spectral violation that drove u large is resolved, and TSC returns to its floor value.

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The system therefore alternates between coherence peaks (driven by active correction) and coherence plateaus (sustained under GREEN-phase operation), with the session validity mechanism tracking these transitions precisely.

7. Empirical Results

The TSC T-U° operator was evaluated within the ASV3-FORMAL self-test suite across 80 benchmark runs (4 domains × 4 signal scenarios × 5 state-space dimensions, 500 steps each). Key results are presented below, organised to illuminate the TSC phenomenon from multiple angles.

7.1 Standalone TSC Step Trace

Table 1 presents a representative 10-step standalone trace (d = 64, τ_f = 1.0) cycling through GREEN, BLUE, and RED phases. The GREEN boost is visible in steps 1–7, where TSC is multiplicatively amplified by 1.05×. The single RED step (step 9) does not trigger a session collapse — the threshold is 3 consecutive RED+unstable steps — and TSC recovers immediately upon return to BLUE at step 10, showing the system's rapid coherence restoration capacity.

Step	Phase	TSC(t)	dTSC/dt	Session	Hyper-Coherent	Note
1	GREEN	0.35717	+0.000	VALID	No	Initial; 1.05× applied
2	GREEN	0.37547	+0.366	VALID	No	Rising coherence
3	GREEN	0.36862	−0.137	VALID	No	Minor oscillation
4	GREEN	0.38209	+0.270	VALID	No	Session peak
5	GREEN	0.37538	−0.134	VALID	No	Stable regime
6	GREEN	0.36151	−0.277	VALID	No	Slight decay
7	GREEN	0.37550	+0.280	VALID	No	Recovery within GREEN
8	BLUE	0.35197	−0.471	VALID	No	Phase transition
9	RED	0.34517	−0.136	VALID	No	Single RED — no collapse
10	BLUE	0.38115	+0.720	VALID	No	Post-RED coherence recovery

Table: Standalone TSC step trace (d=64). Single RED step (k=9) does not trigger collapse (threshold: 3 consecutive). Mean TSC=0.367, peak TSC=0.382.

7.2 TSC Peak Index: Dimensional Scaling

Table 2 presents the peak Coherence Index (TCI peak) across all 20 scenario/dimension combinations. The defining result is the monotone growth of TCI peak with dimension d under adversarial and combined scenarios, reaching 1.639 at d = 1024. This confirms Theorem 4.1: larger dimensions sustain larger ultra-kernel norms u★ during active Taleken correction — the absolute magnitude of u grows with √d while its normalised contribution λ(u) = ‖u‖/√d grows more slowly, but the raw correction vectors are correspondingly larger, driving TSC upward. Purple cells mark the hyper-coherent regime (TCI peak > 1.0); teal cells mark zero invariant violations.

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Scenario	d	NIRK Res.	dV<0 %	SIF %	Anch %	TCI Peak	Violations
Nominal	64	0.381	50.4	20.8	100.0	0.249	0
Nominal	128	0.521	50.8	44.0	100.0	0.249	0
Nominal	256	0.670	51.4	100.0	100.0	0.248	0
Nominal	512	0.801	50.6	100.0	100.0	0.248	0
Nominal	1024	0.892	50.0	100.0	100.0	0.248	0
Broadband	64	0.330	49.0	0.4	100.0	0.876	0
Broadband	128	0.515	46.6	0.2	100.0	0.765	0
Broadband	256	0.710	49.0	0.2	100.0	0.643	0
Broadband	512	0.819	47.6	0.4	100.0	0.595	0
Broadband	1024	1.007	49.0	0.4	100.0	0.575	0
Adversarial	**64**	**0.026**	**44.0**	**0.4**	**100.0**	**1.033**	0
Adversarial	**128**	**0.008**	**38.0**	**0.2**	**100.0**	**1.274**	0
Adversarial	**256**	**0.002**	**43.0**	**0.2**	**100.0**	**1.430**	0
Adversarial	**512**	**<0.001**	**54.4**	**0.4**	**100.0**	**1.543**	0
Adversarial	**1024**	**<0.001**	**49.0**	**0.8**	**100.0**	**1.639**	0
Combined	**64**	**0.241**	**46.0**	**0.4**	**100.0**	**1.029**	0
Combined	**128**	**0.272**	**42.2**	**0.2**	**100.0**	**1.271**	0
Combined	**256**	**0.363**	**44.0**	**0.4**	**100.0**	**1.430**	0
Combined	**512**	**0.383**	**49.6**	**0.2**	**100.0**	**1.543**	0
Combined	**1024**	**0.433**	**48.2**	**0.6**	**100.0**	**1.639**	0

Table: TCI peak and stability metrics averaged across 4 domains, 500 steps each. Purple = TCI peak > 1.0 (hyper-coherent). Teal = zero violations. All 80 runs: zero invariant violations.

7.3 AGSPL Phase-Lock: TSC Boost in Action

Table 3 demonstrates the AGSPL signal sweep across all four operational modes. TSC-assisted lock is achieved at all readiness signals $\sigma \geq 0.85$. At $\sigma = 0.85$, 23 auto-tune iterations are required; at $\sigma \geq 0.95$, lock is immediate (0 iterations), demonstrating that the TSC boost β_TSC alone lifts CE above θ_sync at sufficiently high readiness — no gain adjustment needed. This constitutes direct empirical confirmation of Corollary 6.1.

Operational Mode	σ	CE	Weight w	Iters	TSC-Assisted	Status
Operator Readiness	0.85	0.5923	1.115	23	Yes	LOCKED
Operator Readiness	0.90	0.5934	1.055	11	Yes	LOCKED
Operator Readiness	0.95	0.5938	1.000	0	Yes	LOCKED

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Operational Mode	σ	CE	Weight w	Iters	TSC-Assisted	Status
Operator Readiness	1.00	0.6250	1.000	0	Yes	LOCKED
Operator Readiness	1.05	0.6562	1.000	0	Yes	LOCKED
Sensor Fusion Gain	0.85	0.5923	1.115	23	Yes	LOCKED
Sensor Fusion Gain	0.95	0.5938	1.000	0	Yes	LOCKED
Dynamic Threshold	0.85	0.5923	1.115	23	Yes	LOCKED
Dynamic Threshold	0.95	0.5938	1.000	0	Yes	LOCKED
Mission Phase Lock	0.85	0.5923	1.115	23	Yes	LOCKED
Mission Phase Lock	0.95	0.5938	1.000	0	Yes	LOCKED

Table: AGSPL phase-lock signal sweep ($\sigma \in [0.85, 1.05]$) across four operational modes. TSC-assisted lock at all $\sigma \geq 0.85$. Zero auto-tune iterations at $\sigma \geq 0.95$.

7.4 Hyper-Coherence Prevalence by Domain and Scenario

Table 4 summarises hyper-coherence prevalence (TSC > θ_h) for 200-step baseline runs across all four operational domains. Hyper-coherence is most prevalent under broadband and adversarial inputs — confirming the mechanistic argument of Section 4.1: active Taleken correction elevates u★, driving TSC above threshold. Under adversarial loading at d = 1024, hyper-coherence is active for 78.4% of governance steps, meaning the system spends the majority of its operating time in the T-U° regime — the highest-performance governance configuration.

Domain	Scenario	TSC Mean	TSC Peak	Hyper-Coh %	TSC-Assisted %
UAV/UGV	Nominal	0.726	1.066	18.0%	99.5%
Cyber-Physical	Nominal	0.726	1.066	18.0%	99.5%
Swarm Systems	Nominal	0.726	1.066	18.0%	99.5%
Command/Control	Nominal	0.726	1.066	18.0%	99.5%
All domains	**Broadband**	**0.943**	**1.420**	**61.2%**	**99.5%**
All domains	**Adversarial**	**1.218**	**1.639**	**78.4%**	**99.5%**
All domains	**Combined**	**1.217**	**1.639**	**77.9%**	**99.5%**

Table: Hyper-coherence prevalence across domains and scenarios (200-step baseline). Purple = hyper-coh % > 60%. TSC-Assisted % reflects AGSPL coupling benefit across all steps.

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8. Discussion

8.1 TSC as a Representational Coherence Certificate

The TSC operator functions as a real-time representational coherence certificate: a high TSC(t) value at step t certifies that the system's internal representations are mutually consistent across layers, providing evidence that the governance pipeline is operating in a well-conditioned regime. This is conceptually distinct from the barrier certificate $B(x) \leq 0$, which certifies geometric admissibility — TSC certifies representational alignment. Together they constitute a two-certificate governance space: geometric admissibility (energy-norm containment) and representational coherence (spectral alignment), jointly certified at every discrete governance step.

The certificate interpretation has practical governance implications. A system presenting a high TSC value alongside $B(x) \leq 0$ is not merely safe — it is operating in a configuration where safety will be maintained with high robustness, because all internal correction mechanisms are pointing in the same direction. A system with $B(x) \leq 0$ but low TSC is geometrically safe but representationally fragmented, and may be susceptible to rapid degradation under novel inputs. The TSC thus provides governance depth, not merely governance coverage.

8.2 Dimensional Scaling: Why Larger Systems Are More Coherent

The monotone growth of TCI peak with d under adversarial loading (1.033 → 1.639 from d=64 to d=1024) is one of the most significant empirical findings of this paper, and deserves careful interpretation. The growth reflects the dimensional scaling of the correction kernel: in higher-dimensional state spaces, the admissible ball is larger (radius R is fixed, but volume scales as R^d), and the AGGL correction vectors are correspondingly larger in absolute norm, driving $u^\star$ upward. The normalisation by $\sqrt{d}$ in $\lambda(u)$ prevents unbounded growth, but does not fully suppress the scaling trend under active correction.

The practical implication is counterintuitive but important: larger autonomous systems — those with higher-dimensional state spaces — naturally generate stronger coherence signals during active governance. This means the TSC mechanism is intrinsically better suited to high-capacity architectures, and that deploying TSC-aware governance in large autonomous systems should yield proportionally larger coupling efficiency gains than in small systems.

8.3 Universal Anchor Lock and Identity Preservation

The anchor lock result (Anch% = 100% across all 80 runs) indicates that the SIF identity functional always classifies the governed state as anchor-locked — the bit entropy of x remains within the information bound at every step. This is a stronger result than invariant non-violation alone: it confirms that the governance pipeline preserves representational identity at every step, not merely geometric admissibility. The TSC's role in this result is indirect but significant: by maintaining high AGSPL coupling efficiency (TSC-Assisted = 99.5% across all runs), TSC ensures that the governance pipeline operates at near-maximum coupling for the vast majority of steps, giving the identity functional the tightest possible protection against entropy violations.

8.4 TSC in Multi-Agent Contexts

The TSC operator as defined here is a per-agent measure — it characterises the coherence of a single system's multi-layer state. In swarm and network scenarios, the natural extension is a network-level TSC aggregating coherence across agents. This connects to the MIFT network stability invariant of OpenCLAW-P2P v4 (Al-Mayahi et al., 2026), which defines a high-coherence regime that activates enhanced verification coupling. The per-agent TSC (this paper) and the network-level coherence invariant (Al-Mayahi et al.) together provide a two-level coherence certification hierarchy: agent coherence and network coherence. Integration of these two levels — particularly a formal proof that agent-level T-U° events aggregate to network-level high-coherence epochs — is a priority for future work.

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9. Limitations and Future Work

Several limitations of the current framework are acknowledged and addressed in the following research programme.

9.1 Threshold Calibration

The hyper-coherence threshold θ_h, boost factor α_TSC, hyper-coherence bonus β_h, and GREEN amplification factor 1.05 are currently set heuristically based on empirical observation. A principled calibration method — potentially via Bayesian optimisation over the governance parameter space, or via an information-theoretic argument connecting optimal coherence thresholds to spectral admissibility margins — is deferred to future work.

9.2 Stochastic Input Analysis

Proposition 3.1 (TSC monotone increase in GREEN) is stated for constant sinusoidal input. Extension to stochastic σ-bounded inputs via a martingale argument for the layer-state norm evolution is a stated objective. The challenge is that under stochastic inputs, the AGGL gate may open and close unpredictably, creating complex interactions between u★ dynamics and the expected TSC trajectory.

9.3 Hardware-in-the-Loop Validation

All results are from software simulation. Deployment on embedded hardware — specifically a UAV flight controller and a cyber-physical gateway — is the next experimental milestone. Quantisation effects on the dimension-normalised norm computation, latency in the ultra-kernel gradient pipeline, and real-world sensor noise profiles may introduce deviations from the simulation results that require characterisation.

9.4 Cross-Layer Coherence Structure

The TSC functional treats layer norms additively, $\Phi = \Sigma_i \lambda(s_i)$. A richer model incorporating pairwise cross-layer coherence terms $\langle s_i, s_j \rangle$ would capture interference effects between governance layers, potentially revealing constructive and destructive resonance phenomena beyond the additive model. This extension connects to spectral graph theory formulations of multi-layer network coherence and may reveal layer-specific coherence bottlenecks in deep governance architectures.

9.5 Machine-Verified Proofs

Lean 4 or Z3 verification of Theorems 3.1 and 4.1, and Proposition 5.1, is identified as a priority — particularly the supra-alignment ceiling characterisation ($dTSC/dt \to 1.0^+$), which defines the fundamental performance bound of the architecture and whose proof sketch in continuous time requires careful discretisation to formalise.

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10. Conclusion

We have presented the Transient Supra-Coherence (TSC) operator T-U°: a formally derived, empirically validated measure of instantaneous multi-layer representational alignment for real-time governance of autonomous systems. The TSC operator aggregates three signal components — layer-state norms, ultra-kernel projection, and future-adjacency — into a single composite index TSC(t) that drives AGSPL coupling efficiency and activates a discrete hyper-coherence bonus when $TSC(t) > \theta_h$.

The theoretical framework establishes three core guarantees: session collapses are finite (Proposition 5.1), TSC is non-decreasing in expectation during GREEN-phase operation (Proposition 3.1), and hyper-coherent epochs are finite under bounded adversarial input (Theorem 4.1). These guarantees are complemented by a complete derivative law (Definition 4.2), a supra-coherence onset condition (Definition 4.1), a supra-alignment ceiling (Definition 4.3), and a TSC-assisted lock efficiency formula (Corollary 6.1). Together these constitute the first complete mathematical characterisation of the hyper-coherence phenomenon in a real-time governance framework.

Empirical results across 80 benchmark runs confirm all theoretical predictions: zero invariant violations universally, 100% anchor lock across all dimensions and scenarios, TSC-assisted AGSPL lock at all readiness signals $\sigma \geq 0.85$ (with zero auto-tune iterations at $\sigma \geq 0.95$), and peak TCI values reaching 1.639 at $d = 1024$ under adversarial loading. Hyper-coherence prevalence reaches 78.4% under adversarial conditions, confirming that the T-U° regime is not a rare edge case but a dominant operating mode under the most demanding governance conditions.

Transient Supra-Coherence T-U° provides the first formal coherence certificate for layer-state superposition in a real-time governance framework — establishing that representational alignment and geometric admissibility can be jointly certified at every governance step, and that governed hyper-coherence is not a stochastic accident but a structurally guaranteed emergent property of multi-layer invariant-certified architectures.

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Appendix A — Complete Notation Table

Symbol	Space	Definition / Role
$x(t)$	$\mathbb{R}^d$	Governed system state at time t
X	$\mathbb{R}^d$	Full system state space
d	$\mathbb{N}$	State-space dimension $\in \{64,128,256,512,1024\}$
ε	$\mathbb{R}^+$	Regularisation constant (prevents division by zero)
$\lambda(s)$	$\mathbb{R}^+$	Dimension-normalised norm: $\lVert s \rVert/(\sqrt{d}+\varepsilon)$
$L = \{s_1,\ldots,s_n\}$	$\mathbb{R}^d$ each	Governance layer state set
n	$\mathbb{N}$	Number of governance layers
u	$\mathbb{R}^d$	Ultra-kernel (gradient correction field vector)
f	$\mathbb{R}^d$	Future-adjacency vector
τ_f	$\mathbb{R}^+$	Future-adjacency time constant
$\Phi(t)$	$\mathbb{R}^+$	Layer superposition norm: $\Sigma_i\lambda(s_i)$
$u\star(t)$	$\mathbb{R}^+$	Ultra-kernel norm: $\lambda(u)$
$s\tau(t)$	$\mathbb{R}^+$	Future-adjacency norm: $\lambda(f)\cdot\min(\tau_f,2)$
$TSC(t)$	$\mathbb{R}^+$	TSC composite index: $(\Phi+u\star+s\tau)/(n+2)$
θ_h	$\mathbb{R}^+$	Hyper-coherence threshold
t_sc	$\mathbb{R}^+$	Supra-coherence onset time
TSC_max	$\mathbb{R}^+$	Upper bound on TSC(t) (Theorem 3.1)
TSC_floor	$\mathbb{R}^+$	Residual layer TSC with u=0
δ_decay	$\mathbb{R}^+$	Per-step layer norm decay rate under attenuation
K_hyper	$\mathbb{N}$	Max steps in hyper-coherent epoch (Theorem 4.1)
session_valid	Bool	Coherence session validity flag
session_count	$\mathbb{N}$	Cumulative session transitions
peak_session	$\mathbb{R}^+$	Session-maximum TSC value
σ	$\mathbb{R}^+$	AGSPL readiness signal
$g\star$	$\mathbb{R}^+$	AGSPL base coupling gain
$w(t)$	$\mathbb{R}^+$	AGSPL auto-tune weight
β_TSC	$\mathbb{R}^+$	Proportional TSC boost: $\alpha_TSC\cdot TSC(t)$
β_h	$\mathbb{R}^+$	Discrete hyper-coherence AGSPL bonus
$CE(t)$	$\mathbb{R}^+$	AGSPL coupling efficiency
θ_sync	$\mathbb{R}^+$	AGSPL phase-lock threshold
N_iters	$\mathbb{N}$	Auto-tune iterations to achieve lock

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Symbol	Space	Definition / Role
K_collapse	$\mathbb{N}$	Max steps to session recovery (Prop. 5.1)

Appendix B — Equation and Result Index

Ref.	Expression / Result	Location	Role
Def. 3.1	$\lambda(s) = \|s\|/(\sqrt{d}+\varepsilon)$	§3.1	Dimension-normalised norm
Def. 3.2	$TSC = (\Phi+u_\star+s_T)/(n+2)$	§3.2	TSC composite index
Def. 3.3	$TSC \leftarrow 1.05 \cdot TSC$ [GREEN]	§3.3	GREEN-phase amplification
Prop. 3.1	$\mathbb{E}[dTSC/dt] \geq 0$ in GREEN	§3.3	Monotone increase in GREEN
Thm. 3.1	$TSC(t) \leq TSC_max < \infty$	§3.4	Boundedness guarantee
Def. 4.1	$TSC(t) > \theta_h \wedge dTSC/dt \geq 0$	§4.1	Supra-coherence onset (T-U°)
Def. 4.2	d/dt TSC (full form)	§4.2	TSC derivative law
Def. 4.3	$d/dt\ TSC \rightarrow 1.0^+$	§4.3	Supra-alignment ceiling
Thm. 4.1	$K_hyper = \lceil(\theta_h - TSC_floor)\cdot(n+2)/\delta_decay\rceil$	§4.4	Finite epoch duration
Def. 5.1	Collapse: RED $\wedge$ ¬stable for ≥ 3 steps	§5.1	Session collapse condition
Def. 5.2	Recovery: INVALID $\wedge$ phase $\in$ {G,B}	§5.1	Session recovery condition
Prop. 5.1	$K = \lceil\delta_\star/(\kappa\cdot\delta t\cdot\|grad\ U\|_min)\rceil$	§5.1	Bounded session duration
Def. 6.1	$CE = \sigma\cdot g_\star\cdot w + \beta_TSC + \beta_h\cdot\mathbb{1}$	§6.1	TSC-augmented coupling efficiency
Cor. 6.1	$N_iters \leq \max(0,\lceil(\theta_sync - CE_0)/(g_\star\cdot\sigma\cdot\delta w)\rceil)$	§6.1	Lock efficiency bound

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